

Leveraging Attention for Robust Neural Event Causality Extraction (LEARN-ECE)

RAWDAG, PAIN, TES

Shruti Gupta, Maria Jose Healey, Jasmine Lau

Abstract

The ability to extract event causality from text is crucial to understanding relationships within data, enabling more accurate reasoning, prediction, and decision making in a variety of natural language processing applications, including classification, recommendation, and textual entailment. Event causality extraction (ECE) has many inherent challenges associated with it, such as implicit causality, multiple cause-effect pairs in a sentence, and cause-effect pairs across sentences. This paper explores the integration of causality-aware attention mechanisms into transformer models to address these challenges in event stream data. We investigate several models and evaluated them against MAVEN-ERE to demonstrate the potential of causality-aware attention mechanisms to improve ECE performance and broaden the applicability of causal relations in natural language processing.

1 Introduction

Understanding causal relationships between events in text is a fundamental challenge that underscores the field of natural language processing (NLP), with critical applications such as classification, recommendation systems, event prediction, and textual entailment. Modeling multi-variable event sequences is particularly valuable for identifying patterns and predicting future events, potentially leading to advancements in industries ranging from finance to healthcare. However, extracting event causality from a body of text accurately remains quite challenging due to the natural ambiguity of casual relationships. The challenges posed by language involves issues such as the presence of implicit causality and the complexity of multi-argument events.

Existing methods of extracting events and event sequences from textual corpora often fail to capture the sometimes implicit, oftentimes nuanced,

causal relationships necessary for downstream applications. These challenges posed by the known procedures for extracting events make it difficult¹ to incorporate causality into fields where we know it could be applicable. In this paper, we propose leveraging causality-aware attention mechanisms to enhance the performance of transformer models in conducting ECE. We explore three different approaches (1) a probabilistic attention-to-influence framework that employs an encoder-decoder architecture, (2) a minimum influencing set identification procedure constrained by iterative causal discovery, and (3) a vanilla transformer for event sequences.

Our evaluation is conducted on the annotated MAVEN-ERE dataset from Tsinghua University, which builds off the original MAVEN dataset, but with the addition of event relation extraction (ERE) tasks. The original MAVEN dataset is limited in terms of ERE tasks and it lacks a unified system of annotating², unable to address the challenges of event causality ambiguity and complexity. MAVEN-ERE steps in, to contributing data in hopes of a more accurate and interpretable causal reasoning model for textual data.

2 Background

The ECE task presents several unique challenges. For example, causal relationships often span multiple sentences within a corpus, requiring models to capture both local and global event dynamics. Additionally, the relationships exhibit a many-to-many structure, where a single cause can lead to multiple effects, and an effect can result from multiple causes. These relationships are frequently dispersed over extended spans of text, further complicating ECE and introducing class imbalance issues. Past literature has focused on decomposing ECE

¹<https://proceedings.mlr.press/v202/shou23a.html>

²<https://arxiv.org/abs/2211.07342>

into two subtasks: event extraction and relationship classification.³

To address these challenges, recent research has focused on attention-based architectures capable of capturing both inter- and intra- inferences. For example, Ahmad, Peng, and Chung developed the Graph-Aware Transformer Encoder (GATE), which imposes graph structures into the attention matrix by manipulating the mask matrix so that the attention weights capture pairwise syntactic distances.⁴ Similarly, Shou et al propose an influence-aware attention mechanism by constructing an influence indicator in the decoder to integrate pairwise learning about event dynamics into the model, then combining it with the attention score matrix.⁵

3 Methods

Our experimental design focused on leveraging attention-based frameworks to uncover causal relationships in textual data. The architectures examined include: the Probabilistic Attention-to-Influence (PAIN), Transformer for Event Sequences (TES), and Attention-based Causal-Discovery (ABCD). These frameworks utilize transformer-based architectures, which are known for their flexibility and effectiveness in tasks such as event prediction.

We evaluated the performance of these models using the the MAVEN-ERE dataset, an extensive collection of annotated event relationships.

The PAIN and TES models assess performance using log-likelihood (LL), a metric analogous to logarithmic scoring rules, to predict the next event label. In contrast, RAWDAG evaluates performance using the Structural Hamming Distance, which quantifies the difference between the true and predicted directed acyclic graph (DAG).

3.1 Directed Acyclic Graph Methods (DAG)

In their proposed Attention-Based Causal Discovery (ABCD) algorithm, Rohekar, Gurwicz, and Nisimov further explore the connection between self-attention mechanisms and the nodes of a structural causal model (SCM). The authors consider an SCM to be $X = GX + \lambda U$ where X is a set of latent endogenous random variables, U is the

set of latent exogenous random variables, G represents the causal graph, and λ is a parameter. By rearranging the equation and taking the geometric series expansion, we see that the (i, j) -th element of $(I - G)^{-1}$ corresponds to the sum of the weights of all directed paths from X_j to X_i . Thus, $(I - G)^{-1}$ measures the effect of one variable on another through all directed paths in a causal graph for any value of U .

Similarly, in attention calculations, each column of the value matrix V corresponds to a projection of the input embeddings along a learned vector. The attention matrix A captures the total effect one variable exerts on another for any projection $V(\cdot, j)$ of the input embeddings. Building on this, it can be demonstrated that the SCM covariance, C_x is equivalent to the covariance of the attention output layer, C_z . Under the causal Markov and d-faithfulness assumptions, the causal graph of the SCM can be recovered from the learned weights of the deepest attention layer. In their implementation, Rohekar et al. use an iterative causal discovery (ICD) algorithm, which employs repeated partial correlation conditional independence tests to iteratively expand the size of the potential influence set.

Adapting the ABCD framework to event sequences, we propose Relational Attention Weighting for Directed Acyclic Graphs (RAWDAG). Similar to other methodologies such as GATE and PAIN, RAWDAG seeks to impose structure on the attention matrix to facilitate the learning of causal influence relationships. However, unlike GATE and PAIN, RAWDAG does not rely on externally tagged features to define supplemental structures. Instead, it enables the model to learn an adjustment matrix to modify the attention weights directly.

From the deepest attention layer, we compute an average of the attention weights across heads and initialize a graph based on this weighted attention matrix rather than starting with a fully connected graph as in the ABCD implementation. RAWDAG also incorporates a modified version of the ABCD model’s iterative causal discovery (ICD) algorithm into its training pipeline. We opt to use a perfect oracle during the CI-testing to ensure stability. To encourage sparsity, we adopt Gumbel-Softmax sampling for continuous relaxation during structure learning, adopted from the methodologies in PAIN and TES. The loss is calculated using binary cross-entropy between the learned adjacency matrix of

³<https://www.sciencedirect.com/science/article/pii/S1319157823001477#s0125>

⁴<https://arxiv.org/abs/2010.03009>

⁵<https://proceedings.mlr.press/v202/shou23a.html>

the predicted DAG and the true adjacency matrix. After experimenting, we also modify the loss to encourage sparsity of predictions. This structured approach aims to enhance the interpretability and performance of the model in ECE.

3.2 Probabilistic Attention-to-Influence Neural Model (PAIN)⁶

The Probabilistic Attention-to-Influence Neural Model (PAIN) is designed to analyze temporal event sequences by capturing complex instance-wise interactions between events while identifying the influencer for each event type of interest. PAIN operates on event sequence data and leverages a uniform prior distribution. It employs an attention-to-influence transformation with variational inference to learn an approximate posterior distribution. This allows PAIN to model the conditional likelihood of sequences by sampling from the derived posterior. In essence, PAIN learns an adjacency matrix that informs the attention matrix about longer-term tendencies regarding how one event influences another. This approach ensures that the model focuses attention on the event types that exert the most significant influence on the overall sequence.

Originally for temporal event sequences, PAIN was modified to enhance its performance in causal relation modeling. Some key adjustments include: (1) replacing the original random vector with a binary random matrix that describes pairwise interactions between all event pairs, (2) as a result of the modification of the vector, the variational loss was also changed to better reflect causal relationships, and (3) the prior distribution was defined as: $\mathcal{P}(A) = \prod_{XY} \mathcal{P}(A_{XY} = 1) = 0.2$ for any $X, Y \in \mathcal{L}$, where $\mathcal{L}$ is the label set.

3.3 Transformer for Event Sequences (TES)⁷

The Transformer for Event Sequences (TES) is a vanilla transformer for event sequence modeling, inspired by the Transformer Hawkes Process (THP).

Transformer Hawkes Process (THP): THP uses temporal information, such as timestamps to capture both short- and long-term dependen-

cies between events. The self-attention module of THP can directly select events irrespective of their temporal distance from the current time. In this architecture, each event sequence is fed through embedding layers and N multi-head self-attention modules. The outputs of THP are hidden event representations with historical context.⁸

Transformer for Event Sequences (TES): TES, in contrast, is an autoregressive transformer that models event sequences without time-stamp information. Instead of processing temporal data, TES focuses on categorical event sequences, producing a sequence of categorical probability distributions. The primary components of TES are: (1) event embedding, where the input event subsequence is embedded using position embeddings to, (2) the self-attention layer, which includes multi-head attention modules and a position-wise feed-forward neural network, both of which are connected through residual connections, and (3) attention masking, which ensures that only past events can influence predictions about future events, preserving causal structure. This architecture strives to maximize the log-likelihood of event sequences D , formalized by:

$$\log p_{\theta}(D) = \sum_{k=1}^K \sum_{i=1}^{N_k} \log p_{\theta}^*(l_i)$$

4 Results and Discussion

The MAVEN-ERE dataset is a comprehensive, large-scale human-annotated dataset with improved annotation schemes over the original MAVEN dataset. It contains 103,193 event co reference chains, 1,216,217 temporal relations, 57,992 causal relations, and 15,841 sub-event relations⁹.

Each document in the dataset includes a unique document identifier, a title, content (sentences and tokens), a list of annotated events, and a list of negative instances. The list of annotated events are represented by a unique identifier, event type, event type ID, and mentions. Each mention specifies a unique string ID, the trigger word or phrase, the index of the corresponding sentence, and the offset of the trigger words in the token list¹⁰.

⁶https://proceedings.neurips.cc/paper_files/paper/2023/file/91b047c5f5bd41ef56bfaf4ad0bd19e3-Paper-Conference.pdf

⁷https://proceedings.neurips.cc/paper_files/paper/2023/file/91b047c5f5bd41ef56bfaf4ad0bd19e3-Paper-Conference.pdf

⁸<https://arxiv.org/abs/2002.09291>

⁹<https://aclanthology.org/2022.emnlp-main.60.pdf>

¹⁰<https://github.com/THU-KEG/MAVEN-dataset/tree/main>

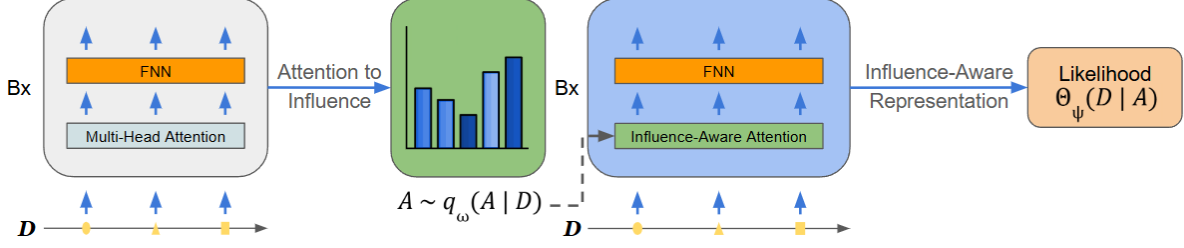


Figure 1: Architecture of the Probabilistic Attention-to-Influence Neural model where D are events sequences, and A is a binary random matrix.

4.1 Data Preprocessing

For the PAIN and TES models, we preprocessed a subset of the dataset to create event streams, which are represented as a list of dictionaries. Each dictionary contains information about the time since the start of the event, the time since the last event, and the type of event. Each document contained multiple sentences and could span up to more than 500 words, so event stream extraction helped address the length issues and filter out noise. This preprocessing resulted in 2,913 documents, 570 event labels, and 2,828 event streams. We then split the data into train, validation, and test sets in a 60 : 20 : 20 ratio for evaluation. Consequently, we ended up working with 2913 documents, 570 event labels, and 2828 event streams. Following the preliminary processing, the data was split into train-dev-test sets (60/20/20) % for evaluation.

Dataset	Docs	Labels	Streams
MAVEN-ERE	2,913	570	2,828

Table 1: Dataset Summary: # of documents, # of event labels, and # of event streams

For the RAWDAG model, preprocessing differed slightly, as the intent was to construct a DAG rather than predict event sequences, and the transitive causal relations automatically filled by the MAVEN-ERE system were removed. Each event (node) was averaged over its tokens so that it corresponded to a single input token.

4.2 RAWDAG Model Results

The RAWDAG baseline, using 4 attention heads and 4 attention layers, achieved a loss of approximately 0.13, and increasing the number of attention layers to 8 reduced the loss to 0.10. However, the Structural Hamming Distances (SHD) revealed discrepancies between the true and predicted DAGs.

Most errors were due to the over-prediction of children compared to true number of children for the node sets, although, the model did not predict edges in the wrong direction.

To address this issue of over-prediction, we introduced a sparsity parameter to the loss function to penalize dense graph predictions. By inserting this parameter decreased the average SHD from 255 to 65 without significantly increasing the amount of missing edges and maintaining no edges of in the wrong direction. Additionally, the threshold for CI test was tuned to limit the number of candidate node pairs. After tuning, we found that setting a higher threshold helps the density problems. Further adjusting the number of attention heads and increasing the number of epochs did not yield any significant improvement to the adjusted loss. This behavior remained consistent across training and test sets.

4.3 PAIN and TES Model Results

The PAIN and TES models were evaluated on the MAVEN-ERE dataset focusing on predicting the next event labels.

The PAIN model focuses on the local dynamics for each label of interest within an event sequence. Its primary objective is to predict whether a specific event of interest will reoccur based on historical data. In contrast, the TES model focuses on capturing the per-event log-likelihood by using an unaltered history representation, where the event labels are modeled using a multinomial distribution.

The results for next-event prediction log-likelihood are presented in Figures 2 and 3. PAIN maximized the negative log-likelihood to -1.0063 , whereas TES maximized the log-likelihood to -0.909 . While TES outperformed PAIN during the validation process, there were several contributing factors. Likely due to differences in training time and computational resource allocation. PAIN's

training was constrained by GPU limitations, leading to fewer epochs and suboptimal hyperparameter tuning compared to TES. TES did not require as many resources as PAIN, leading us to conduct more training with TES.

The hyperparameters used for training each model that resulted in the best log-likelihood are shown in Table 2. It is important to note that final parameter selection for both models was lead by the minimum loss observed on the validation set during evaluation.

5 Conclusion

Refining attention mechanisms is a powerful approach to extracting event relationships in the context of both event pairs and event sequences. Imposing probabilistic or graph based structures on the attention matrix allows transformer-based architectures to learn relationships based on directed influence and improve in the event causality extraction task. Future avenues of research might aim to refine the causal discovery process by employing GCN layers. Additionally, more work is needed to generalize the models discussed in this paper to work with full segments or documents beyond event streams.

6 References

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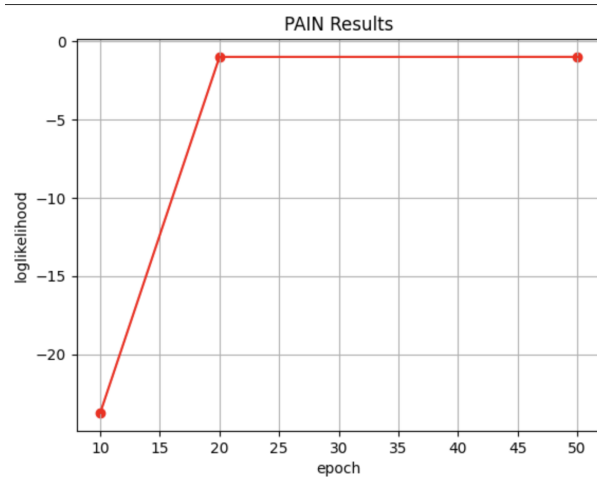


Figure 2: PAIN

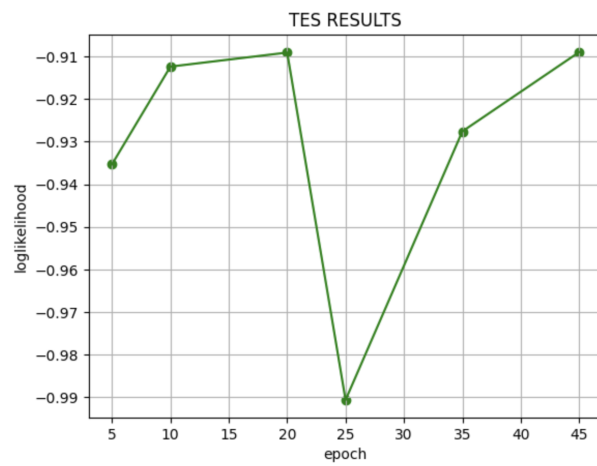


Figure 3: TES

Parameter	RAWDAG	PAIN	TES
epoch	5	50	45
batch_size	8	16	16
d_model	128	128	512
d_inner	96	128	256
d_v	128	64	32
d_k	128	64	128
n_head	4	4	8
n_layers	8	6	8
dropout	0.3	0.376	0.26242
learning_rate	0.0001	0.00008	0.00013
num_samples	-	5	-
threshold	-	0.44995	-

Table 2: Hyperparameters